Elementary Problems
Sequence

Version 1.0

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Basic Work

1. Let $(a_n)_{n\geqslant 1}$ be an arithmetic sequence with $a_3 = 20$ and $a_{12} = 200$. Find the leading term a_1.

2. Let $(a_n)_{n\geqslant 1}$ be an arithmetic sequence with leading term $a_1 = 1$ and common difference 3. Find the sum S_n of the first n terms of the sequence.

3. Find the general formula of the following.

   (a) $1^2 + 2^2 + \cdots + n^2$;

   (b) $1^3 + 2^3 + \cdots + n^3$;

   (c) $1 \times 2 + 2 \times 3 + \cdots + n(n+1)$.

4. Let $(a_n)_{n\geqslant 1}$ be a geometric sequence with $a_2 = 2$ and $a_4 = 4$. Find a_8.

5. Let $(a_n)_{n\geqslant 1}$ be a geometric sequence with leading term $a_1 = 1$ and common ratio 3. Find the sum S_n of the first n terms of the sequence.

6. Let $(a_n)_{n\geqslant 1}$ be a geometric sequence with leading term $a_1 = 1$ and common ratio $\dfrac{1}{3}$. Find the infinite sum $a_1 + a_2 + \cdots$.

7. Find all infinite sequence $(a_n)_{n\geqslant 1}$ which is both an arithmetic sequence and a geometric sequence.

8. Find all positive real numbers $a \leqslant b \leqslant c \leqslant d \leqslant e$ such that both a, b, c and c, d, e are arithmetic sequences, while both a, c, e and b, c, d are geometric sequences.

9. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_n = \dfrac{n}{2^n}$ for $n \geqslant 1$. Find the infinite sum $a_1 + a_2 + \cdots$.

10. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_n = \dfrac{1}{(3n-2)(3n+1)}$ for $n \geqslant 1$. Find the sum S_n of the first n terms of the sequence.

11. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_n = \dfrac{1}{n(n+1)(n+2)}$ for $n \geqslant 1$. Find the sum S_n of the first n terms of the sequence.

12. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_n = \dfrac{n}{n^4 + n^2 + 1}$ for $n \geqslant 1$. Find the sum S_n of the first n terms of the sequence.

13. (IMO shortlist 1975) Let $a_1, a_2, \ldots, a_n, \ldots$ be a sequence of real numbers such that $0 \leqslant a_n \leqslant 1$ and $a_n - 2a_{n+1} + a_{n+2} \geqslant 0$ for $n = 1, 2, 3, \ldots$. Prove that

$$0 \leqslant (n+1)(a_n - a_{n+1}) \leqslant 2 \quad \text{for } n = 1, 2, 3, \ldots.$$

14. Prove or disprove the following for an infinite sequence of real numbers (or positive integers).

    (a) An arithmetic sequence is unbounded.

    (b) A non-constant geometric sequence is unbounded.

    (c) A monotonic sequence is bounded.

    (d) A periodic sequence is bounded.

    (e) A recurrence sequence (a_n) defined by $a_{n+2} = \alpha a_{n+1} + \beta a_n$ for some nonzero constants α, β is unbounded.

    (f) A bounded sequence is periodic.

    (g) A bounded sequence has infinitely many equal terms.

    (h) A bounded sequence has a largest term (and a smallest term).

    (i) A sequence with a largest term (or a smallest term) has an upper bound (or a lower bound).

    (j) If the terms a_n of a sequence are arbitrarily close to 0 for sufficiently large n, then the partial sums $a_1 + a_2 + \cdots + a_n$ are bounded.

    (k) If the partial sums $a_1 + a_2 + \cdots + a_n$ of a sequence (a_n) are bounded, then the terms a_n are arbitrarily close to 0 for sufficiently large n.

<u>Recurrence Sequence</u>

15. (IMO shortlist 1969) Let $a_0, a_1, a_2, \ldots$ be determined with $a_0 = 0$, $a_{n+1} = 2a_n + 2^n$. Prove that if n is a power of 2, then so is a_n.

16. (CSMO 2008 Problem 2) Suppose the sequence $\{a_n\}$ satisfies

$$a_1 = 1,\ a_{n+1} = 2a_n + n \cdot (1 + 2^n),\ n = 1, 2, 3, \ldots.$$

    Find the general formula of a_n.

17. Let $(a_n)_{n \geqslant 1}$ be the sequence defined by $a_1 = 1$ and

$$a_n = 2(a_1 + a_2 + \cdots + a_{n-1})$$

    for $n \geqslant 2$. Find the general formula for a_n.

18. Let $(a_n)_{n \geqslant 0}$ be the sequence defined by $a_0 = 2$, $a_1 = 4$ and

$$a_{n+2} = 3a_{n+1} - 2a_n + 2 \times 3^n$$

    for $n \geqslant 0$. Find the general formula for a_n.

19. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_1 = 5$ and

$$a_{n+1} = \frac{3a_n - 4}{a_n - 1}$$

for $n \geqslant 1$. Find the general formula for a_n.

20. (CWMO 2004 Problem 5) Suppose the sequence $\{a_n\}$ satisfies $a_1 = a_2 = 1$, and

$$a_{n+2} = \frac{1}{a_{n+1}} + a_n,\ n = 1, 2, \ldots.$$

Find a_{2004}.

21. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_1 = 2$ and $a_n = a_1a_2\cdots a_{n-1} + 1$ for $n \geqslant 2$. Find the value of

$$a_1a_2\cdots a_n\Big(1 - \frac{1}{a_1} - \frac{1}{a_2} - \cdots - \frac{1}{a_n}\Big).$$

22. (CWMO 2008 Problem 1) The sequence $\{a_n\}$ of real numbers satisfies $a_0 \neq 0, 1$, $a_1 = 1 - a_0$, $a_{n+1} = 1 - a_n(1 - a_n)$ for $n = 1, 2, \ldots$. Prove that for any positive integer n, we have

$$a_0a_1\cdots a_n\Big(\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_n}\Big) = 1.$$

23. (CWMO 2001 Problem 1) Suppose the sequence $\{x_n\}$ satisfies $x_1 = \dfrac{1}{2}$, $x_{n+1} = x_n + \dfrac{x_n^2}{n^2}$. Prove that $x_{2001} < 1001$.

24. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_1 = \dfrac{1}{2}$ and $a_{n+1} = \dfrac{2n-1}{2n+2} \cdot a_n$ for $n \geqslant 1$. Prove that $a_1 + a_2 + \cdots + a_n < 1$ for any $n \geqslant 1$.

25. (IMO shortlist 1984) Let c be a positive integer. The sequence $\{f_n\}$ is defined as follows:

$$f_1 = 1,\ f_2 = c,\ f_{n+1} = 2f_n - f_{n-1} + 2 \quad (n \geqslant 2).$$

Show that for each $k \in \mathbb{N}$ there exists $r \in \mathbb{N}$ such that $f_kf_{k+1} = f_r$.

26. (CSMO 2004 Problem 3)

    (1) Does there exist an infinite sequence $\{a_n\}$ of positive integers such that for any positive integer n, we have $a_{n+1}^2 \geqslant 2a_na_{n+2}$?

    (2) Does there exist an infinite sequence $\{a_n\}$ of positive rational numbers such that for any positive integer n, we have $a_{n+1}^2 \geqslant 2a_na_{n+2}$?

27. Suppose the sequence $(a_n)_{n\geqslant 1}$ of positive real numbers satisfies $a_{n+1} = \sqrt{a_n + 2}$ for $n \geqslant 1$. Prove that (a_n) is a monotonic sequence.

28. Let $(F_n)_{n\geqslant 0}$ be the sequence defined by $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for $n \geqslant 0$. Find the value of $F_{n+1}^2 - F_n F_{n+2}$.

29. Let $(a_n)_{n\geqslant 1}$ be the sequence defined by $a_1 = a_2 = 1$ and

$$a_{n+2} = \frac{a_{n+1}^2 + 2}{a_n}$$

for $n \geqslant 1$. Prove that a_n is an integer for any $n \geqslant 1$.

30. (IMO 1976 Problem 6) A sequence (u_n) is defined by

$$u_0 = 2,\ u_1 = \frac{5}{2},\ u_{n+1} = u_n(u_{n-1}^2 - 2) - u_1 \quad \text{for } n = 1, 2, \ldots.$$

Prove that for any positive integer n we have

$$\lfloor u_n \rfloor = 2^{\frac{2^n - (-1)^n}{3}}.$$

31. (CSMO 2011 Problem 7) Suppose the sequence $\{a_n\}$ satisfies

$$a_1 = a_2 = 1,\ a_n = 7a_{n-1} - a_{n-2},\ n \geqslant 3.$$

Prove that for each $n \in \mathbb{N}$, $a_n + a_{n+1} + 2$ is a perfect square.

32. Let $(a_n)_{n\geqslant 0}$ be the sequence defined by $a_0 = 1$, $a_1 = 3$ and $a_{n+2} = 3a_{n+1} - a_n$ for $n \geqslant 0$. Prove that $a_n a_{n+1} + 1$ is a perfect square for any $n \geqslant 0$.

33. (CWMO 2003 Problem 5) Suppose the sequence $\{a_n\}$ satisfies $a_0 = 0$,

$$a_{n+1} = ka_n + \sqrt{(k^2 - 1)a_n^2 + 1},\ n = 0, 1, 2, \ldots,$$

where k is a given positive integer. Prove that each term of the sequence $\{a_n\}$ is an integer, and $2k \mid a_{2n}$ for $n = 0, 1, 2, \ldots$.

34. (IMO shortlist 1981) A sequence (a_n) is defined by means of the recursion

$$a_1 = 1,\ a_{n+1} = \frac{1 + 4a_n + \sqrt{1 + 24a_n}}{16}.$$

Find an explicit formula for a_n.

Record

v1.0: problems 1–34